

Lecture 12: review 2

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Important learning objectives (not all!)

Conceptual understanding

- ▶ Energy minimization governs bond formation
- ▶ Quantum coupling creates bonding/antibonding splitting
- ▶ Band gaps arise from Bragg reflection at zone boundaries
- ▶ Band filling determines conductivity
- ▶ Effective mass captures band curvature effects
- ▶ Lattice defects dictate τ

Quantitative skills

- ▶ Derive density of states $Z(E)$ for 3D, 2D, and 1D
- ▶ Calculate Fermi energy from electron density
- ▶ Compute probabilities from Fermi-Dirac
- ▶ Compute carrier concentrations in semiconductors
- ▶ Use charge neutrality, mass action law, and E_F

Energy minimization and bond formation

Main reason for bonding

Bonding occurs because the bonded state has lower energy than isolated atoms ($\Delta E < 0$)

- ▶ Total energy: repulsive + attractive terms
- ▶ Equilibrium at $dE/dr = 0$ (force balance $F = 0$)
- ▶ Bond energy E_c = depth of minimum at r_0
- ▶ Curvature determines stiffness and bulk modulus

$$E(r) = \frac{A}{r^n} - \frac{B}{r^m} \quad (1)$$

(2)

Common error: Confusing equilibrium ($dE/dr = 0$) with stability ($d^2E/dr^2 > 0$)

Bonding-antibonding splitting and bond types

Quantum mechanical origin

- ▶ Coupling splits degenerate levels:

$$E = E_0 \pm A$$

- ▶ **Bonding state** ($E_0 - A$): symmetric, lower energy
- ▶ **Antibonding state** ($E_0 + A$): antisymmetric, higher energy

Four bond types

- ▶ **Ionic**: electron transfer (NaCl)
- ▶ **Metallic**: delocalized e^- (Cu)
- ▶ **Covalent**: shared pairs (Si)
- ▶ **van der Waals**: weak dipole (Ar)

Property trends

Property	Ionic	Metallic	Covalent	vdW
Stiffness	High	Variable	High	Low
Conductivity	Low	High	Variable	Low

Checkpoint: bond formation

Quick check

For a bond described by $E(r) = A/r^{12} - B/r^6$:

1. Which term is repulsive? Which is attractive?
2. What condition determines the equilibrium separation r_0 ?
3. If A increases, does r_0 increase or decrease?

Checkpoint: bond formation

Quick check

For a bond described by $E(r) = A/r^{12} - B/r^6$:

1. Which term is repulsive? Which is attractive?
2. What condition determines the equilibrium separation r_0 ?
3. If A increases, does r_0 increase or decrease?

Answers:

1. A/r^{12} is repulsive (positive, short-range); $-B/r^6$ is attractive
2. $dE/dr = 0$ at $r = r_0$
3. $r_0 = (2A/B)^{1/6}$ increases with A (stronger repulsion $\rightarrow$ larger separation)

Free electron model

Main assumptions of the model

1. Independent electrons (no e^- - e^- interaction)
2. Jellium (uniform positive background)
3. Infinite potential walls

Infinite well approximation

- ▶ Work function $W \sim 4\text{--}5$ eV
- ▶ Thermal energy $k_B T \approx 0.025$ eV at 300 K
- ▶ $W \gg k_B T \Rightarrow$ electrons confined

Energy in a 3D well

$$E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

with $n_x, n_y, n_z = 1, 2, 3, \dots$

Level spacing for $L = 1$ cm:

$$\Delta E \sim \frac{h^2}{8mL^2} \approx 10^{-15} \text{ eV}$$

Nearly continuous!

Critical failure: Cannot explain insulators or semiconductors (no band gaps)

3D Density of States (DOS) derivation: $Z(E) \propto E^{1/2}$

Geometric construction

1. Define $R^2 = n_x^2 + n_y^2 + n_z^2$
2. Constant E surfaces are spheres:
 $R = \sqrt{8mL^2E/h^2}$
3. Count states with energy $\leq E$:

$$N(E) = \underbrace{\frac{1}{8}}_{\text{octant}} \times \underbrace{\frac{4\pi}{3}R^3}_{\text{sphere}} \times \underbrace{2}_{\text{spin}} \quad (3)$$

Density of States

$$Z(E) = \frac{dN}{dE} = \frac{4\pi L^3(2m)^{3/2}}{h^3} \sqrt{E}$$

Important assumptions:

- ▶ Factor 1/8: $n_x, n_y, n_z > 0$ only
- ▶ Factor 2: spin degeneracy
- ▶ $E^{1/2}$: 3D geometry

Common mistake: Confusing $N(E)$ (integrated) with $Z(E)$ (density)

Fermi-Dirac statistics and Fermi energy

Fermi-Dirac distribution

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

Key properties

- ▶ $f(E_F) = 1/2$ at *all* temperatures
- ▶ At $T = 0$: step function
- ▶ At $T \rightarrow \infty$: constant at $1/2$

Fermi energy

At $T = 0$, all states below E_F filled:

$$N = \int_0^{E_F} Z(E) dE$$

Solving:

Fermi Energy

$$E_F = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3}$$

Depends *only* on electron density!

Emission mechanisms: thermionic vs field emission

Thermionic Emission

- ▶ Thermal activation over barrier
- ▶ Requires $E > E_F + W$
- ▶ Temperature-dependent:

$$J \propto \exp\left(-\frac{W}{k_B T}\right)$$

- ▶ Arrhenius behavior

Field Emission

- ▶ Quantum tunneling through barrier
- ▶ High field $\sim 10^9$ V/m required
- ▶ Temperature-independent
- ▶ “Cold cathode” emission

Photoelectric effect



$$KE_{\max} = \hbar\omega - W$$

- ▶ Photon energy must exceed work function: $\hbar\omega > W$

Bloch theorem and periodic potentials

Definition of a periodic potential

$V(x + a) = V(x)$ where a is the lattice constant

Bloch theorem

$$\psi_k(x) = u_k(x)e^{ikx}$$

where $u_k(x + a) = u_k(x)$

Key properties

- ▶ $u_k(x)$ is periodic with lattice
- ▶ e^{ikx} describes propagation
- ▶ Translation: $\psi_k(x + a) = e^{ika}\psi_k(x)$

Physical interpretation

- ▶ Probability density is periodic:

$$|\psi_k(x + a)|^2 = |\psi_k(x)|^2$$

- ▶ All unit cells are equivalent
- ▶ First Brillouin zone is enough:

$$-\frac{\pi}{a} \leq k \leq \frac{\pi}{a}$$

Kronig-Penney model and band gap formation

Dispersion Relation

$$\cos(ka) = P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a)$$

where $P = \frac{ma}{\hbar^2} V_0 w$ and $\alpha = \sqrt{2mE/\hbar^2}$

Band gap origin

- ▶ Constraint: $|\cos(ka)| \leq 1$
- ▶ Gaps where $|\text{RHS}| > 1$
- ▶ Gaps at $k = n\pi/a$ (zone boundaries)

You should know: explain gaps using Bragg reflection $\rightarrow$ standing waves $\rightarrow$ energy splitting

Intuition on band gap

At $k = n\pi/a$: Bragg condition

$$2a = n\lambda$$

Standing waves form with different energies depending on charge distribution relative to ions

Checkpoint: band theory

Quick Check

1. Why must $|\cos(ka)| \leq 1$ in the Kronig-Penney model?
2. At what k -values do band gaps occur?
3. What physical phenomenon causes the gap?

Checkpoint: band theory

Quick Check

1. Why must $|\cos(ka)| \leq 1$ in the Kronig-Penney model?
2. At what k -values do band gaps occur?
3. What physical phenomenon causes the gap?

Answers:

1. Because $\cos(ka)$ is bounded by ± 1 for real k ; when the right-hand side of the equation exceeds this, no propagating solution exists
2. $k = n\pi/a$ (Brillouin zone boundaries)
3. Bragg reflection creates standing waves; waves with maxima at ions vs. between ions have different energies

Effective mass: from band curvature to dynamics

Derivation

- ▶ Group velocity: $v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k}$
- ▶ Crystal momentum: $\hbar \frac{dk}{dt} = F$
- ▶ Acceleration:

$$a = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2} F$$

Effective mass

$$m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1}$$

Interpretation

- ▶ Large curvature $\rightarrow$ small m^* (light)
- ▶ Flat band $\rightarrow$ large m^* (heavy)
- ▶ **Negative m^*** near VB max $\rightarrow$ holes!

Free electron: $E = \hbar^2 k^2 / (2m)$ gives $m^* = m$ exactly

Band filling and conductivity

Why full bands don't conduct?

1. Velocity symmetry: $\sum_k v_k = 0$
2. No available states for redistribution

Band filling and materials

- ▶ **Metal:** E_F crosses band
- ▶ **Semiconductor:** small gap (~ 1 eV)
- ▶ **Insulator:** large gap (> 3 eV)

Fermi level

Fermi level position determines conductivity:

- ▶ E_F in band $\rightarrow$ metal
- ▶ E_F in gap $\rightarrow$ semiconductor/insulator

You should know: Classify materials from band diagrams by checking E_F position and gap size

Semiconductor carrier concentration

Boltzmann approximation

Valid when E_F in gap ($E - E_F \gg k_B T$):

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right)$$

Electron concentration

$$N_e = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

$$N_c = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2}$$

Hole concentration

$$N_h = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

$$N_v = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2}$$

Common error: Wrong sign in exponential or forgetting $T^{3/2}$ prefactor

Intrinsic semiconductors

Intrinsic condition

$N_e = N_h = N_i$ (equal electrons and holes, no doping)

Arrhenius plot

Intrinsic carrier concentration

$$N_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$$

- ▶ Activation energy is $E_g/2$ (half gap!)
- ▶ Exponential dominates over $T^{3/2}$

$\ln(N_i)$ vs $1/T$

Slope = $-E_g/(2k_B)$

Example:

- ▶ Si: $E_g = 1.1$ eV
- ▶ Ge: $E_g = 0.66$ eV
- ▶ Ge has higher N_i at same T

Opposite to metals: Semiconductor conductivity *increases* with temperature

Doping: donors and acceptors

Donors (n-type)

- ▶ Group V in Si: P, As, Sb
- ▶ Extra valence electron
- ▶ Level E_D below E_c (~ 10 – 50 meV)

Hydrogen-like model

$$E_D = \frac{m_e^*}{m_0} \frac{1}{\epsilon^2} \times 13.6 \text{ eV}$$

Acceptors (p-type)

- ▶ Group III in Si: B, Al, Ga
- ▶ One fewer electron
- ▶ Level E_A above E_v (~ 10 – 50 meV)

Si example

- ▶ $\epsilon \approx 12$, $m_e^* \approx 0.3m_0$
- ▶ $E_D \approx 30$ meV (shallow)
- ▶ $k_B T \approx 25$ meV at 300 K
- ▶ Most dopants ionized!

Charge neutrality and mass action law

Charge neutrality

$$N_e + N_A^- = N_h + N_D^+$$

Crystal must be electrically neutral

Charge carriers:

- ▶ Positive: holes (N_h), ionized donors (N_D^+)
- ▶ Negative: electrons (N_e), ionized acceptors (N_A^-)

Mass action law

$$N_e \cdot N_h = N_i^2$$

Holds at equilibrium regardless of doping!

Limiting cases:

- ▶ n-type ($N_D \gg N_A, N_i$): $N_e \approx N_D$
- ▶ p-type ($N_A \gg N_D, N_i$): $N_h \approx N_A$

Workflow for semiconductor doping problems

(1) Identify majority carrier from doping → (2) Use charge neutrality → (3) Use mass action for minority

Worked example: doped silicon calculation

Problem

Si at 300 K with $N_D = 10^{16} \text{ cm}^{-3}$ donors. Given: $N_i = 10^{10} \text{ cm}^{-3}$, $N_c = 2.8 \times 10^{19} \text{ cm}^{-3}$. Find N_e , N_h , and E_F .

Step 1: Identify type and majority carrier

► $N_D = 10^{16} \gg N_i = 10^{10} \Rightarrow$ n-type, electrons are majority

Step 2: Apply charge neutrality

$$N_e \approx N_D = 10^{16} \text{ cm}^{-3}$$

Step 3: Apply mass action law for minority carrier

$$N_h = \frac{N_i^2}{N_e} = \frac{(10^{10})^2}{10^{16}} = 10^4 \text{ cm}^{-3}$$

Step 4: Calculate Fermi level

$$E_F = E_c - k_B T \ln \left(\frac{N_c}{N_e} \right) = E_c - (0.026) \ln \left(\frac{2.8 \times 10^{19}}{10^{16}} \right) = E_c - 0.21 \text{ eV}$$

Checkpoint: semiconductor calculations

Quick verification

For p-type Si with $N_A = 10^{17} \text{ cm}^{-3}$ and $N_i = 10^{10} \text{ cm}^{-3}$:

1. What is the majority carrier concentration?
2. What is the minority carrier concentration?
3. Is E_F closer to E_c or E_v ?

Checkpoint: semiconductor calculations

Quick verification

For p-type Si with $N_A = 10^{17} \text{ cm}^{-3}$ and $N_i = 10^{10} \text{ cm}^{-3}$:

1. What is the majority carrier concentration?
2. What is the minority carrier concentration?
3. Is E_F closer to E_c or E_v ?

Answers:

1. $N_h \approx N_A = 10^{17} \text{ cm}^{-3}$ (holes are majority)
2. $N_e = N_i^2 / N_h = 10^{20} / 10^{17} = 10^3 \text{ cm}^{-3}$
3. E_F closer to E_v (p-type shifts Fermi level toward valence band)

Fermi level position and doping

Intrinsic semiconductor

$$E_F = \frac{E_g}{2} + \frac{3}{4} k_B T \ln \left(\frac{m_h^*}{m_e^*} \right)$$

Near mid-gap if $m_h^* \approx m_e^*$

n-type doping

$$E_F = E_c - k_B T \ln \left(\frac{N_c}{N_e} \right)$$

Higher doping $\rightarrow E_F$ closer to E_c

p-type doping

$$E_F = E_v + k_B T \ln \left(\frac{N_v}{N_h} \right)$$

Higher doping $\rightarrow E_F$ closer to E_v

Degenerate doping

Very high doping can push E_F into the band (metallic behavior)

Direct vs indirect band gap semiconductors

Direct gap (GaAs, InP)

- ▶ CB min and VB max at same k
- ▶ Vertical transitions: $\Delta k \approx 0$
- ▶ Photon momentum negligible
- ▶ First-order process
- ▶ **High efficiency** for LEDs/lasers

$$\hbar\omega = E_g$$

Indirect gap (Si, Ge)

- ▶ CB min and VB max at different k
- ▶ Requires lattice scattering for Δk
- ▶ Second-order process (lower probability)

$$\hbar\omega_{\text{photon}} + \hbar\omega_{\text{lattice}} \approx \hbar\omega_{\text{photon}} = E_g$$

Checkpoint: advanced concepts

Quick check

1. Why does the mass action law $N_e \cdot N_h = N_i^2$ hold regardless of doping?
2. If you add donors to an intrinsic semiconductor, what happens to the hole concentration?

Checkpoint: advanced concepts

Quick check

1. Why does the mass action law $N_e \cdot N_h = N_i^2$ hold regardless of doping?
2. If you add donors to an intrinsic semiconductor, what happens to the hole concentration?

Answers:

1. It's a thermodynamic equilibrium constraint; the product of carrier concentrations is fixed by temperature and band gap
2. Hole concentration *decreases*: $N_h = N_i^2 / N_e$ and N_e increases

More conceptual connections

Key things to remember

- ▶ Bond formation $\leftrightarrow$ energy minimization
- ▶ DOS: $1/8$ (octant) $\times 2$ (spin) $\times E^{1/2}$ (3D)
- ▶ $F(E_F) = 1/2$ always
- ▶ Gaps form at $k = n\pi/a$ because of Bragg reflection
- ▶ Negative m^* $\leftrightarrow$ hole behavior

Tips to solve problems

1. Identify what is given/unknown
2. Interpret the problem
3. Choose appropriate equations
4. Check units carefully
5. “does this answer make sense?”

Practice problems

Here are a few practice problems from the previous lectures — make sure you understand each of them!

Conceptual questions: density of states derivation (Lec. 07)

Test your understanding:

1. Why do we only count the positive octant in n -space?
▷ *Hint: What are the allowed values of n_x, n_y, n_z ?*
2. Where does the factor of 2 for spin enter the derivation?
▷ *Hint: How many electrons can occupy each spatial state?*
3. Explain physically why $Z(E) \propto E^{1/2}$ in 3D but not in 1D or 2D
▷ *Hint: How does the “surface area” in n -space scale with dimension?*

Conceptual questions: Fermi-Dirac statistics (Lec. 07)

Test your understanding:

Q1: Why is $F(E_F) = \frac{1}{2}$ regardless of temperature?

▷ *Hint: Substitute $E = E_F$ into the Fermi-Dirac formula*

Q2: At $T = 0$, a metal has N electrons. Where do they reside in energy space?

▷ *Hint: What does the step function tell you?*

Q3: If electron density doubles, by what factor does E_F change?

▷ *Hint: $E_F \propto (N/V)^{2/3}$*

Conceptual questions: free electron model and Fermi-Dirac statistics (Lec. 08)

- ▶ **Q1:** At $T = 0$, all states below E_F are filled. If you suddenly doubled the electron density N/V without changing anything else, would E_F double? Justify quantitatively.
- ▶ **Q2:** The Fermi temperature of copper is $T_F \approx 80,000$ K, yet copper melts at ~ 1400 K. What does this imply about how many electrons near E_F are thermally excited at room temperature?
- ▶ **Q3:** In the photoelectric effect, increasing light *intensity* increases photocurrent but not $K_{\max}$. Explain why this is natural in the photon picture but paradoxical classically.

Conceptual questions: periodic potentials and band structure (Lec. 08)

Conceptual questions:

- ▶ **Q1:** In the Kronig-Penney dispersion relation, solutions require $\left| P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) \right| \leq 1$. What happens physically to an electron whose energy falls in a region where this inequality is violated?
- ▶ **Q2:** Band gaps open at $k = n\pi/a$ because of the Bragg condition. If you could continuously “turn off” the lattice potential ($V_0 \rightarrow 0$), what would happen to the gaps and the dispersion relation?
- ▶ **Q3:** The effective mass $m^* = \hbar^2 (\partial^2 E / \partial k^2)^{-1}$ can be negative near the top of a band. Does a negative effective mass violate Newton’s laws? What is the lattice’s role?
- ▶ **Q4:** Consider two hypothetical 1D crystals with the same lattice constant a but different barrier strengths V_0 . Which one has wider band gaps? Which one has lighter effective masses near band edges? Can both be true simultaneously?

Conceptual questions: band structures (Lec. 10)

Test your understanding:

1. **Q1:** Why does a full band contribute zero current even with an applied field?
2. **Q2:** Why does a hole have a negative effective mass?
3. **Q3:** Can a semiconductor conduct at 0 K? Justify this using the Fermi-Dirac distribution
4. **Q4:** Where is the Fermi level in a metal?
5. **Q5:** What if $|m_h^*| \ll |m_e^*|$? What do you expect for the carrier that dominates the conduction of this system?

Conceptual questions: band structures (Lec. 10)

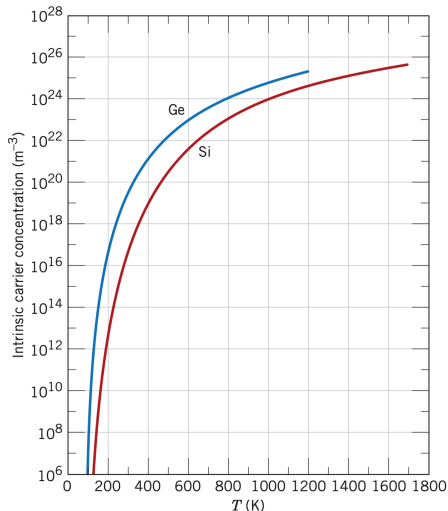
Reflect and discuss:

1. **Q1:** Do you expect the concentration of carriers to change with T in metals?
2. **Q2:** Why do we have to multiply the density of states $Z(E)$ by the Fermi-Dirac distribution $f(E)$ to get the carrier concentration?
3. **Q3:** What are the “engineering knobs” that we have available to change the conductivity of a material?

Conceptual questions: N vs T (Lec. 11)

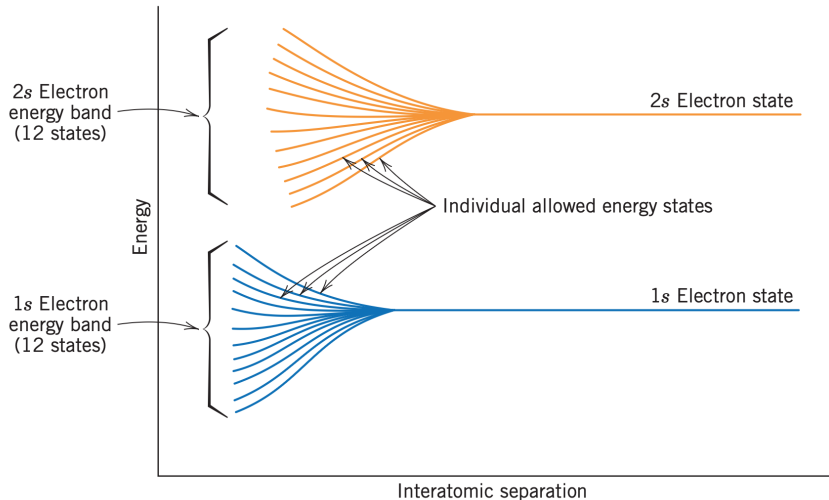
Q: Using what you learned in MAT SCI 120, explain the diagram on the right.

- ▶ Why is the curve behaving like it is?
- ▶ What explains the trends?
- ▶ Why the concentration for Ge is higher than Si?
- ▶ How would this curve look like for GaAs?



Conceptual questions (Lec. 11)

Q: Using what you learned in MAT SCI 120, explain the following diagram. What does

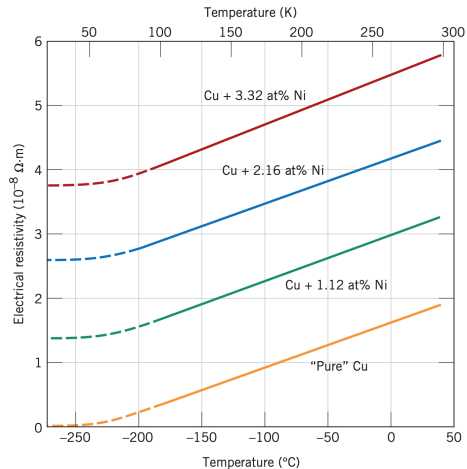


it mean?

Conceptual questions (Lec. 11)

Q: Using what you learned in MAT SCI 120, explain the diagram on the right.

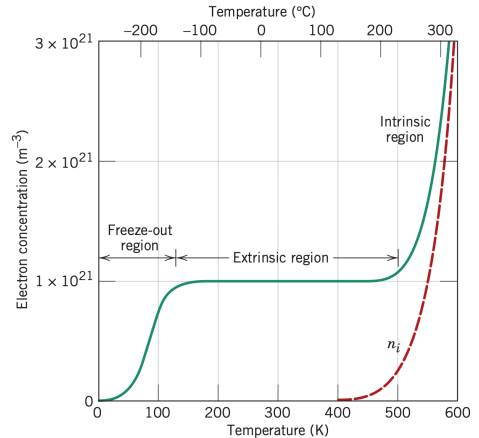
- ▶ What is the *physical meaning* of each regime/feature?
- ▶ What *mechanism* explains the trend(s)?
- ▶ Use the Drude model ($\sigma = N_e e^2 \tau / m$) and/or the wave picture (band structure etc.) to justify your explanation.
- ▶ Identify which parameter(s) change most with temperature for this material: N_e , τ , m^* and why.



Conceptual questions (Lec. 11)

Q: Using what you learned in MAT SCI 120, explain the diagram on the right.

- ▶ What is the *physical meaning* of each regime/feature?
- ▶ What *mechanism* explains the trend(s)?



Conceptual questions (Lec. 11)

1. Two semiconductors have identical τ and identical donor density, but different m^* . Which has higher σ and justify using both *mobility* and *DOS* arguments?
2. Why adding donors as dopants increases N_e but *reduces* N_h (even though donors do not directly “remove holes”)?
3. The 3D DOS near a band edge scales as $Z(E) \propto (m^*)^{3/2} \sqrt{E - E_c}$. If m^* increases, what happens to:
 - ▶ the number of available states near E_c ,
 - ▶ mobility (qualitatively),
 - ▶ and the net conductivity at fixed E_F ?